STA2002 Tutorial 1

1. Let $X \sim \mathcal{N}(\mu, \sigma^2)$ for some $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}^+$. Please show that the moment generating function M_X of X is given by:

$$M_X(t) = \exp\left(\mu t + \frac{1}{2}\sigma^2 t^2\right)$$

By definition,

$$M_x(t) = E\left(e^{xt}\right) = \int e^{xt} \frac{1}{\sqrt{(2\pi\sigma^2)}} e^{-\frac{1}{2}(x-\mu)^2/\sigma^2} dx$$

Define

$$z = \frac{x-\mu}{\sigma}, \quad \text{which implies} \quad x = z\sigma + \mu.$$

Then, using the change-of-variable technique, we get

$$\begin{aligned}
M_x(t) &= e^{\mu t} \int e^{z\sigma t} \frac{1}{\sqrt{(2\pi\sigma^2)}} e^{-\frac{1}{2}z^2} \left|\frac{dx}{dz}\right| dz \\
&= e^{\mu t} \int e^{z\sigma t} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \\
&= e^{\mu t} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2 - 2\sigma t + (\sigma t)^2)} e^{1/2(\sigma t)^2} dz \\
&= e^{\mu t} e^{\frac{1}{2}(\sigma t)^2} \int \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z-\sigma t)^2} dz \\
&= e^{\mu t} e^{\frac{1}{2}\sigma^2 t^2} \\
&= \exp\left\{\mu t + \sigma^2 t^2/2\right\}
\end{aligned}$$

2. If the moment-generating function of X is given by $M(t) = e^{500t + 5000t^2}$, find

$$P\left[27060 \leq (X - 500)^2 \leq 50240\right]$$

From the last question, we know that:
A random variable X has a normal distribution with mean μ and variance σ^2, written as $X \sim N\left(\mu, \sigma^2\right)$, its moment generating function (mgf) is given by

$$M(t) = \mathbb{E}\left(e^{tX}\right) = e^{\mu t + \frac{1}{2}\sigma^2 t^2}, \quad t \in \mathbb{R}.$$

The mean and variance of X are

$$\mathbb{E}(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

Also, recall that: If $X \sim N\left(\mu, \sigma^2\right)$, then $Z^2 = (X-\mu)^2/\sigma^2 \sim \chi^2(1)$ (Theorem 3.3-2 in book).
Then we have, $X \sim N\left(500, 100^2\right)$, so $(X - 500)^2/100^2 \sim \chi^2(1)$ and

$$P\left[2.706 \leq \left(\frac{X - 500}{100}\right)^2 \leq 5.024\right] = 0.975 - 0.900 = 0.075$$

3. Let $X \sim Gamma(\alpha, \beta)$, derive the variance of X by two ways: direct computation and computing through moment generating function.

By the definition of the Gamma distribution, X has probability density function:

$$f(x) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Way1:
From last tutorial, we know that $E(X) = \frac{\alpha}{\beta}$. Following the same logic, we have

$$\begin{aligned} E(X^2) &= \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^{\infty} x^2 \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^{\infty} x^{\alpha+1} e^{-\beta x} dx \\ &= \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha+2)}{\beta^{\alpha+2}} \int_0^{\infty} \frac{\beta^{\alpha+2}}{\Gamma(\alpha+2)} x^{\alpha+1} e^{-\beta x} dx \\ &= \frac{\Gamma(\alpha+2)}{\Gamma(\alpha)\beta^2} \\ &= \frac{\alpha(\alpha+1)}{\beta^2} \end{aligned}$$

$$Var(X) = E(X^2) - E^2(X) = \frac{\alpha(\alpha+1)}{\beta^2} - \frac{\alpha^2}{\beta^2} = \frac{\alpha}{\beta^2}$$

Way2:

$$M_X(s) = \int_{x=0}^{\infty} e^{sx} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = \frac{\beta^\alpha}{(\beta-s)^\alpha} \int_{x=0}^{\infty} \frac{(\beta-s)^\alpha}{\Gamma(a)} x^{\alpha-1} e^{-(\beta-s)x} dx = \frac{\beta^\alpha}{(\beta-s)^\alpha} = (1-\frac{s}{\beta})^{-\alpha}, \quad s < \beta.$$

$$M'_X(s) = \frac{\beta^\alpha \alpha}{(\beta-s)^{\alpha+1}} \text{ and } M''_X(s) = \frac{\beta^\alpha \alpha(\alpha+1)}{(\beta-s)^{\alpha+2}}$$

Hence setting $s = 0$:

$$\begin{aligned} M'_X(0) &= \frac{\beta^\alpha \alpha}{(\beta-0)^{\alpha+1}} = \frac{\beta^\alpha \alpha}{\beta^{\alpha+1}} = \frac{\alpha}{\beta} \\ M''_X(0) &= \frac{\beta^\alpha \alpha(\alpha+1)}{(\beta-0)^{\alpha+2}} = \frac{\alpha(\alpha+1)}{\beta^2} \\ Var(X) &= E(X^2) - E^2(X) = \frac{\alpha(\alpha+1)}{\beta^2} - \frac{\alpha^2}{\beta^2} = \frac{\alpha}{\beta^2} \end{aligned}$$

Note:

$$\int_{-\infty}^{\infty} f(x)dx = \int_0^{\infty} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x} dx = 1 \text{ if } \beta > 0, \alpha \geq 0.$$

This is a very useful formula when working with the Gamma distribution.

4. A candy maker produces mints that have weight of 20.4 grams. Assume that the distribution weights of these mints is $N(21.37,\ 0.4^2)$

   (a) Let X denote the weight of a single mint selected at random from the production line. Find $P(X < 20.857)$

   (b) Let $\bar{X}$ equal the sample mean of the 100 mints selected and weighed on a particular shift. Find $P(21.31 \leq \bar{X} \leq 21.39)$

(a)

$$P(X < 20.857) = P\left(\frac{X - 21.37}{0.4} < \frac{20.857 - 21.37}{0.4}\right) = P(Z < -1.282) = 1 - P(Z \leq 1.282) = 0.10$$

(b)

$$P(21.31 \leq \bar{X} \leq 21.39) \approx P\left(\frac{21.31 - 21.37}{0.4/10} \leq Z \leq \frac{21.39 - 21.37}{0.4/10}\right) = P(-1.50 \leq Z \leq 0.50) = 0.6915 - 1 + 0.9332 = 0.6247$$